

Scalability & Future Plan

Date	3 July 2026
Team ID	SWTID-2026-2260
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses SQLite database, which is suitable for small-scale applications only.	Limited scalability for large datasets and concurrent users.	High
2	Single-machine Flask deployment.	Performance may decrease with increasing user requests.	High
3	Basic authentication and email verification.	Requires stronger security for production environments.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited simultaneous users on a local server.	Deploy the application on IBM Cloud or AWS using load balancing and scalable cloud infrastructure.
2	Data Storage	Uses SQLite database.	Upgrade to MySQL or PostgreSQL for improved performance and scalability.
3	Performance	Predictions are processed on a single server.	Optimize model inference, implement caching, and use cloud-based deployment for faster response times.

4	Security	Basic form validation and Gmail SMTP authentication.	Implement user authentication, password encryption, HTTPS, role-based access control, and secure API communication.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Deploy the application on IBM Cloud with online accessibility.	Next 2 Months	Enables access from anywhere and supports multiple users
Phase 2	Integrate advanced Machine Learning and Deep Learning models for higher prediction accuracy.	Next 4 Months	Improves prediction performance and decision-making accuracy.
Phase 3	Add user authentication, loan application history, and an administrative dashboard.	Next 6 Months	Enhances security, user management, and application monitoring.
Phase 4	Integrate banking APIs for real-time loan processing and automated verification.	Future Release	Provides seamless banking integration and real-world deployment capabilities.